

Fahad Anwar

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EDUCATION

Inderkum High School

Sacramento, CA

GPA | W: 4.8, UW: 4.0

Class of 2026

- **Coursework:** AP Calculus AB (5), AP Computer Science A (5), AP Physics C Electricity and Magnetism (4), IB Math AA HL (IP), IB Computer Science HL (IP), AP Calculus BC (IP)

University of California, San Diego

La Jolla, CA

Bachelor of Science in Electrical Engineering, Minor in Cognitive Science

Class of 2030

SAT: 1530 (750E, 780M)

EXPERIENCE

MIT Intro. to Technology, Engineering, & Science (MITES) Semester | Scholar July 2025 - December 2025

- Selected as 1 of 240 students globally from a pool of 5,100+ applicants (4.7% acceptance rate) to complete rigorous, college-level STEM coursework: Neuroscience and Science Communication
- Co-developed a Bayesian predictive ML model in Python (PyMC) to flag early-onset schizophrenia risk, incorporating 8 literature-weighted clinical and behavioral markers (e.g., CHR flagging OR $\approx$ 85, family history OR=7.0) and achieving an AUC of 0.96 across 16 training chains
- Wrote an award-winning research article on CRISPR-Cas9 and its future implications under the mentorship of Doudna Lab researcher Seyone Chithrananda and Science Communication instructor Asenette Ruiz.

Incoming Amazon Hardware Engineer | Intern

May 2027 - August 2027

- Awarded the \$40,000 Amazon Future Engineer (AFE) Scholarship, along with a guaranteed Software Engineering Internship at Amazon, in recognition of exceptional academic achievement

Independent Physics Research at CSU Sacramento | Scholar

June 2025 - July 2025

- Conducted a self-directed experimental study at CSU Sacramento in collaboration with physics professor Brianna Santangelo, investigating how inductance affects resonance frequency in series RLC circuits
- Designed and assembled circuit configurations from scratch, collected and analyzed quantitative and qualitative data trends, and authored a 4,000-word research paper documenting findings

TECHNICAL PROJECTS

Home Automation System (Arduino, ESP32, C#, Python)

July 2025 - Present

- Engineered a multi-room home automation system using an Arduino UNO R4 and multiple ESP32 microcontrollers, integrating DHT22 temperature/humidity sensors, PIR motion detectors, servo motors, and moisture sensors
- Automated bedroom climate control (humidifier/fan activation), motion-triggered LED lighting, app-controlled window blinds, a scheduled home sprinkler system, and a custom PVC drip-irrigation system for a vegetable garden; controlled via localhost and Arduino IoT Cloud

4-Bit DAC (R-2R Ladder, Op-Amp, Arduino)

April 2026 - Present

- Designed and built a 4-bit digital-to-analog converter using an R-2R resistor ladder network driven by an Arduino microcontroller, converting 4-bit binary inputs to discrete analog voltage levels
- Integrated an op-amp output stage for signal buffering and gain control; validated all 16 output states with a multimeter, confirming linearity and expected step resolution across the full input range

LANA — Linear Algebraic Nodal Analysis (Python, MATLAB, Circuit Prototyping)

May 2026 - Present

- Implemented the LANA algorithm to solve nodal voltage problems in DC resistive circuits by constructing nonsingular linear systems ($Ax = b$) and solving them programmatically in Python/MATLAB
- Physically built and measured multi-resistor circuits with DC voltage and current sources on a breadboard, then verified computed node voltages against measured values to validate the mathematical model

SKILLS & INTERESTS

Languages: English, Urdu, Punjabi

Skills: Python, PyMC, C#, Java, MATLAB, Arduino, ESP32, CAD, Communication

Interests: Coding, Research, Electronics, Networking, Wrestling, and Classic Literature